Star.txt-files

for file in infile

Stars_List
[Time, lambda, Flux]

Changing lambda to velocities using:

$$V = c \cdot \frac{\lambda - \lambda_0}{\lambda_0}$$

Find stars with orbiting planets by using MatPlotLib and looking at curve, the curve is steep over time

Use by-eye measurments to estimate V_{max} and period P, and get a rough estimated mass of planets.

Check if planet eclipses star by looking at flux-curve.

Estimate radius and density if possible

Finding more accurate values:

ranges are found by looking at speed curve →

V_r max_range = [lowest possible, highest possible]

P_range = [—— " ——]

t_0_range = [—— " ——]

Least square method:

```
for i in Vr_range:
    for j in P_range:
        for k in t0_range:
```

$$diff = \sum_t \left(V_r^{data}(t) - V_r^{Model}(t, t_0, P, V_r)\right)^2$$

Lowest value for "diff" gives most accurate V_r, P and t_0. Use this to calculate more accurate Mass of Planet.

Plotting absorbtion lines:

```
def F(lambda, lambda0, sigma, Fmin, Fcont):
    return Fcont + (Fmin - Fcont) e^(-(λ-λ0)^2 / 2σ^2)
```

First data point:

$F(\lambda, \lambda_0(t=0), \sigma, F_{min}, F_{cont})$

Later data point:

$F(\lambda, \lambda_0(t-P/4), \sigma, F_{min}, F_{cont})$

Finding σ:

$$\sigma = \frac{FWHM}{\sqrt{8 \ln 2}}$$

$$FWHM = \left(\frac{2\lambda_0}{c}\right)\sqrt{\frac{2kT \ln 2}{m}}$$